

LEONARD AARONS-DITSON

AI RESEARCH & SYSTEMS ENGINEER

Model Behaviour · Agent Evals · Tool Use · Scientific Computing

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UCL-trained quantum-information physicist with commercial data-science experience, now doing empirical research on AI agents, evals and tool use. The common thread is measurement: does a score, a passing test suite or a valid schema capture what it should?

EDUCATION & RESEARCH FOUNDATION

University College London · PGCert Physics with Distinction, 2021 (within the MSc Physics programme) · BSc Physics, 2018

Selected postgraduate marks: **Advanced Quantum Theory 93.5%** · **Modelling Quantum Many-Body Systems 86%**.

Postgraduate Quantum-Information Research · UCL (Prof. Sougato Bose's group)

- ▶ Built Python/NumPy and Qiskit simulations of spin-chain communication, density-matrix evolution and open-system dynamics, with parameter sweeps and validation against expected dynamics; custom-gate circuits, one run on IBM Quantum hardware.
- ▶ Delivered a UCL lecture on quantum simulators to students, alumni and academic staff; wrote a quantum-computing article commended for making a difficult subject clear.

INDEPENDENT AI RESEARCH & ENGINEERING | 2023-PRESENT

How do agentic systems behave when tools, evaluators, execution environments and protocol choices become part of the experiment?

Agentic Physics Bench · controlled tool-use evaluation of Claude via Claude Code CLI

github.com/retinapeg/agentic-physics-bench

When a model has deterministic computation available, when does it actually choose to use it?

- ▶ Ran 162 scored episodes on held-out physics tasks under no-tool, optional-tool and required-tool conditions, with frozen tasks and deterministic grading. Optional-tool use went from 0/12 (V1) to 54/54 (V2) while accuracy stayed at ceiling.
- ▶ **Accuracy alone did not measure the behaviour of interest**, so tool choice is now scored as its own outcome.
- ▶ **The harness itself can contaminate results**. Trace audits found a second, undeclared model being consulted in 31/48 development calls; per-call controls now invalidate any such episode before it is scored.

Coding-Agent Cross-Review · multi-agent software evaluation on real code

github.com/retinapeg/agent-workflow-orchestrator

Can an independent coding agent catch a defect that another agent and its own acceptance tests miss?

- ▶ Claude Code implements and Codex CLI reviews in isolated Git worktrees, with schema-validated reviews and fresh test gates. In two live runs the reviewer caught a Unicode lowercasing defect that four passing acceptance tests missed; Claude fixed it on revision.
- ▶ **A green test suite did not mean the code was correct**. Independent review added information the acceptance criteria did not capture, but it only counts as evidence when environments, schemas and gates are controlled well enough to audit.

Protocol Reliability in Agent Evals · harness and failure analysis

github.com/retinapeg/institutional-workbench

Are apparent differences in agent behaviour properties of the model, or artefacts of the protocol around it?

- ▶ Ran a frozen 96-call role-prompting pilot on two CLI systems under three prompt conditions: 48/48 vs 15/48 schema-valid calls by system, with 33/96 lost to malformed output, timeouts and provider errors.
- ▶ **Protocol reliability dominated the apparent model and role signal**. A model ranking on this data would have measured the harness, so the failure became the finding: validate the measurement pipeline before comparing models.

Grounded Agent Planning · structured-output planning with deterministic fact validation

Private repository

- ▶ Built a Codex CLI strict-JSON-Schema planning pipeline whose deterministic validators rejected unsupported claims in 151/251 first-pass plans (60%) across 259 runs on 257 real job listings, forcing evidence-linked replanning.
- ▶ **Schema validity is not factual grounding**. Evidence checks plus replanning turned unsupported generation into an auditable workflow.

COMMERCIAL EXPERIENCE

Data Scientist · Sticky · London

2020-2021

- ▶ Built Python/ML analyses and product-placement decision tools over imperfect customer, payment and product data, surfacing commercial drivers that contributed to a **27% increase in profitability**; investigated SaaS support incidents and translated findings into recommendations for non-technical colleagues and investor-facing discussions.

Further experience: 999 Emergency Call Handler, East of England Ambulance Service NHS Trust (Jan-Jun 2026) · Medical Laboratory Assistant, Imperial College Healthcare NHS Trust (Sep 2023-Nov 2024).

Career context: major cancer treatment interrupted conventional career progression; since recovery, returned to frontline NHS operations and intensive independent AI research and engineering.

METHODS & TOOLS

AI research & evals: experiment design, harness design, tool-use studies, trace analysis, structured outputs (JSON Schema/Pydantic), prompt/protocol design, deterministic validation, failure analysis, multi-agent orchestration; Claude Code, Codex CLI, LLM APIs.

Engineering: Python, NumPy, pandas, SciPy, scikit-learn, Jupyter, pytest, Git/GitHub (worktrees), GitHub Actions CI, REST/JSON APIs, OAuth 2.0, SQL/PostgreSQL (Neon), DuckDB, Streamlit, C/C++, OpenCL/PyOpenCL (GPU kernels), Qiskit, Qutip; Excel, Power BI.

SELECTED DISTINCTIONS

- ▶ Selected as **1 of 15 UCL Physics students** for Cumberland Lodge; met HM Queen Elizabeth II and HRH Prince Philip.
- ▶ Competitive UCL rower (crew won Cambridge Head); organised a 72-hour rowathon that raised **£25,000 for Cancer Research UK**.